Zero inflated models

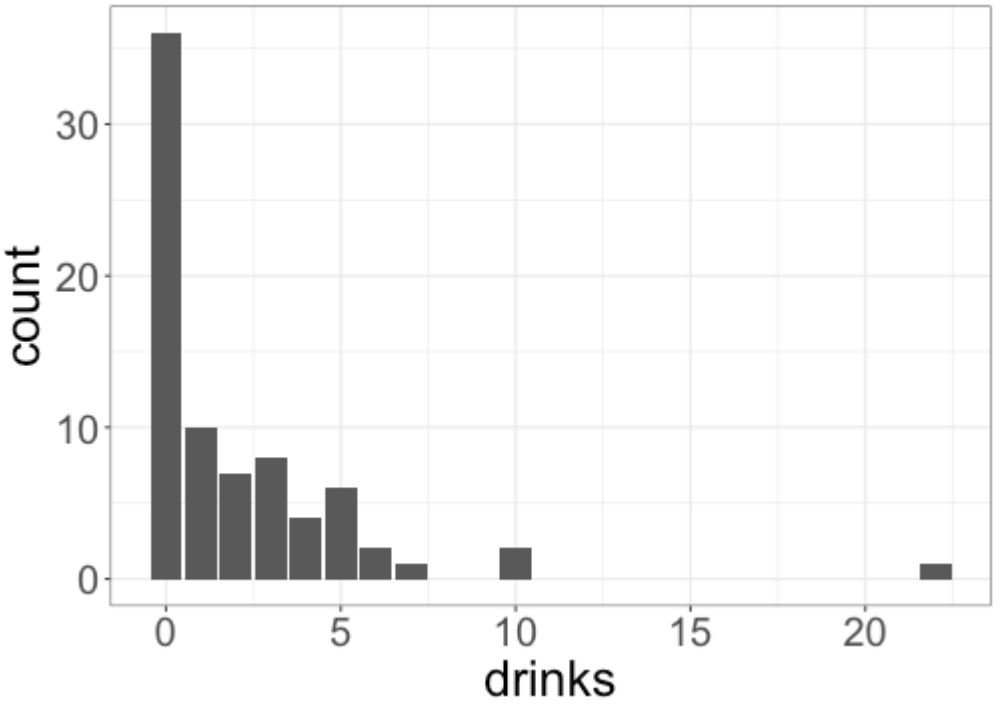
Last time: College drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: the number of drinks the student reports consuming
- `sex`: an indicator for whether the student identifies as male
- `OffCampus`: an indicator for whether the student lives off campus
- `FirstYear`: an indicator for whether the student is a first-year student

> Our goal: model the number of drinks students report consuming.

EDA: drinks



Excess zeros

never drink (always 0): "structural 0s"

sometimes drink (sometimes 0): "random 0s"

The problem:

- There are two groups of people contributing 0s to the data: those who never drink, and those who sometimes drink but didn't drink last weekend
- By itself, a Poisson distribution doesn't do a good job modeling data that is a mixture of these two groups

Imagine latent variable Z_i

$Z_i = 1 \rightarrow$ never drinks $(Y_i \equiv 0)$

$Z_i = 0 \rightarrow$ sometimes drinks $(Y_i \sim \text{Poisson})$

$Z_i \sim \text{Bernoulli}(\alpha_i)$

$Y_i | Z_i = 1 = 0$

$Y_i | Z_i = 0 \sim \text{Poisson}(\lambda_i)$

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 FirstYear_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 OffCampus_i + \beta_2 Male_i$$

> This is called a *mixture* model (it is a mixture of two different models). We *can* fit this model on the observed data (we don't need to observe Z_i)

$\alpha_i = P(Z_i = 1)$ = P(student i is a never drinker)

Zero-inflated Poisson (ZIP) model

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$Y_i | Z_i = 0 \sim Poisson(\lambda_i)$

λ_i = average # of drinks for a student who sometimes drinks (given particular values for explanatory variables)

What do α_i and λ_i represent?

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 FirstYear_i$$

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> What do α_i and λ_i represent?

α_i = probability the student doesn't drink, λ_i = average number of drinks if the student *does* drink

Class activity

Work on the class activity (handout).

Class activity: The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\ OffCampus_i + 1.02\ Male_i$$

> What is the estimated probability that a first year student never drinks?

$$\hat{\alpha}_i = \frac{e^{-0.60 + 1.14}}{1 + e^{-0.60 + 1.14}} \approx 0.63$$

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\ OffCampus_i + 1.02\ Male_i$$

> What is the estimated average number of drinks for a male student who lives off campus and sometimes drinks?

$\hat{\lambda}_i = \exp\{0.75 + 0.42 + 1.02\} \approx 8.43$

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\ OffCampus_i + 1.02\ Male_i$$

> What is the estimated probability that a male first year student who lives off campus had at least one drink last weekend?

$P(Y_i \geq 1) = 1 - P(Y_i = 0)$

$P(Y_i = 0) = e^{-\hat{\lambda}_i}(1-\hat{\alpha}_i) + \hat{\alpha}_i$

$\Rightarrow \hat{P}(Y_i \geq 1) = 1 - (e^{-8.93}(1-0.63) + 0.63) \approx 0.37$

Another example: going fishing

Data on 250 groups of visitors to a state park. For each group, we have:

- `count`: the number of fish caught by the group
- `camper`: whether the group brought a camper van (0 = no, 1 = yes)
- `child`: the number of children in the group
- `persons`: the total number of people in the group
- `LOS`: length of stay (in days)

> Why might the number of fish have excess zeros? What would be the two latent groups in the data?

going fishing vs. act fishing

Going fishing

We propose the following model:

$$P(\text{Fish}_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 \text{Camper}_i + \gamma_2 \text{Children}_i + \gamma_3 \text{Persons}_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Camper}_i + \beta_2 \text{Children}_i + \beta_3 \text{Persons}_i$$

> Is there any problem with comparing λ_i (the expected number of fish caught for those who *do* go fishing) between groups?

Offsets

We will account for the length of stay by including an **offset** in the model:

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Camper}_i + \beta_2 \text{Children}_i + \beta_3 \text{Persons}_i + \log(\text{LOS}_i)$$

← offset (no β associated w/ offset)

$$\log(\lambda_i) - \log(\text{LOS}_i) = \beta_0 + \beta_1 \text{Camper}_i + \cdots + \beta_3 \text{Persons}_i$$

$$\log\left(\frac{\lambda_i}{\text{LOS}_i}\right) = \beta_0 + \beta_1 \text{Camper}_i + \cdots + \beta_3 \text{Persons}_i$$

↑ rate of fish caught (average # caught per day)

⇒ Can now interpret βs in terms of rates

Motivation for offsets

We will account for the length of stay by including an **offset** in the model:

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Camper}_i + \beta_2 \text{Children}_i + \beta_3 \text{Persons}_i + \log(\text{LOS}_i)$$

response: still # of fish caught

(random component is still the same)

only change is interpretation of βs

Interpretation

$$\log(\widehat{\lambda}_i) = -1.13 + 0.0175\text{Camper}_i - 0.564\text{Children}_i + 0.451\text{Persons}_i + \log(\text{LOS}_i)$$

e.g. $\hat{\beta}_1 = 0.0175$ $\exp\{0.0175\} \approx 1.018$

having a camper van is associated with an estimated increase of 1.8% in # of fish caught per day, after accounting for other variables (children & persons), for the groups that do go fishing

Weekend drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

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> Should we include an offset for this model?

When to use offsets

Offsets are useful in count regression (Poisson, negative binomial, etc.) when our counts come from groups of very different sizes (e.g., different lengths of stay). The offset lets us interpret model coefficients in terms of rates instead of raw counts.

> With your neighbor, brainstorm some other data scenarios where our response is a count variable, and an offset would be useful. What would our offset be?

- goals scored in soccer
  offset: # minutes played
  # games played
- competitive eating (# hotdogs)
  offset: # minutes
- # car crashes (offset: city population)
- # of sales of product (offset: time on shelf)

Class activity

Work on the class activity (handout).